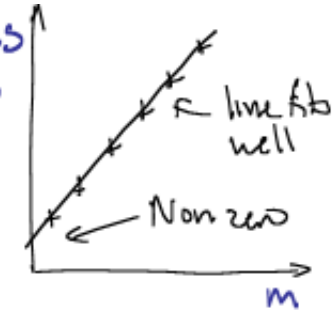
P1 May 2009 TZ1

1.

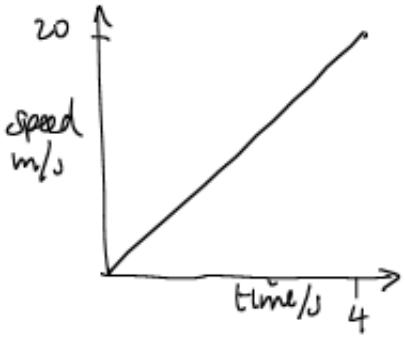
1. Expect that mams so intercept should be zero. Non zero intercept ⇒ systematic error



Line passes very close to points ⇒ small random error A

2.

2. Distance = Area = ½ × 20 × 4 = 40m B



3.

3. This is a tricky one! I'd say PE was being converted to heat but it not an option. The elevator actually has a counter weight so when it goes down the weight is lifted so no change in PE. The electric motor moves the elevator so electric energy → Thermal D

Hmm.

4.

4. If orbit is circular then there must be a centripetal force acting towards the centre. This is due to gravity



B.

5.

5. If the Earth pulls the student down then the student must pull the Earth up.



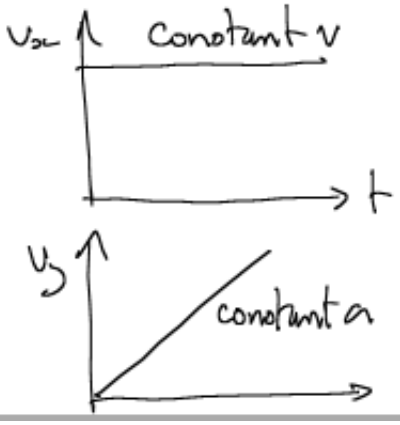
D

Note: A lot of students think that the Normal force on the student is the answer but then both forces act on the same body. It is Newt. 1st that says these are equal.

6.

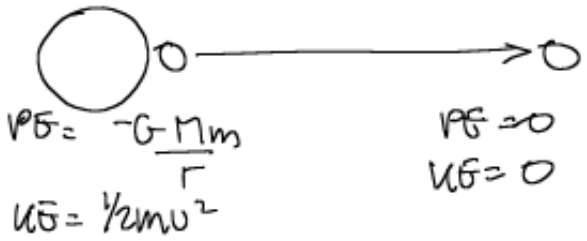
6. This is projectile motion ⇒ parabolic.

A.



7.

7.



Loss of KE = gain of PE

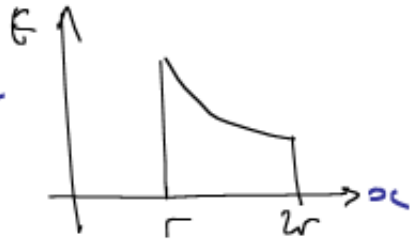
$\frac{1}{2}mv^2 = \frac{GMm}{r}$

A

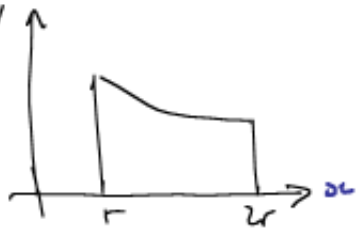
8.

8. Work done = area under Fac graph

E - Force/charge



so Area = Work/charge of Fac



(Area of E-ac) × q = Work done

A

9.

9. $\frac{\text{thermal capacity}}{\text{specific heat capacity}} = \frac{Q/\Delta T}{Q/m\Delta T}$

⇒ units are kg

D

10.

10. In kinetic model atoms are perfectly elastic, small spheres with no forces between them unless colliding.

⇒ force not always zero
PE of molecules = zero (const)
KE change all the time
momentum changes all the time

⇒ B

11.

11. Work is done ⇒ volume increases
∴ not isochoric

Internal E increases ⇒ Temp increases ∴ not isothermal

Heat added ⇒ not adiabatic

⇒ must be isobaric B

12.

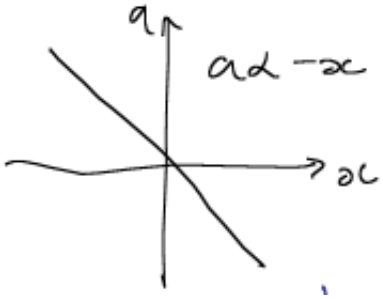
12. When water freezes the entropy of the water gets less.
2nd Law ⇒ entropy of the universe must increase

B

13.

13. Defn of SHM is that accel is proportional to displacement from a fixed point and directed towards that point

A



14.

14. The time period is not dependent on amplitude so this is the same, T

$E = \frac{1}{2} k x^2$ so if x is ×2

PE = ×4, 4E

A

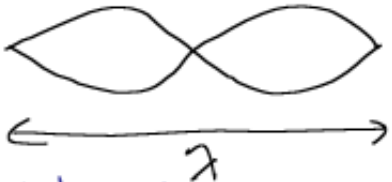
15.

15. slight reduction in amplitude

⇒ lightly damped C

16.

16.



λ = 2× distance between nodes

B

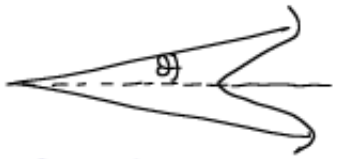
17.

17. Eeeeeoowww
High → Low

So frequency varies during the journey

D

18.

18. 

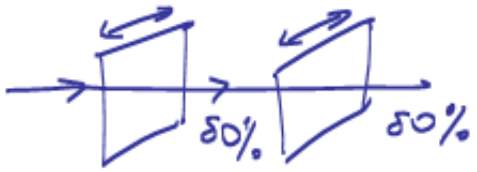
$b \sin\theta = \lambda$

but if θ is small $\sin\theta = \theta$ (radians)

$\Rightarrow b\theta = \lambda \quad \theta = \lambda/b$

width of max^{m} $= 2\theta = 2\lambda/b$ A

19.

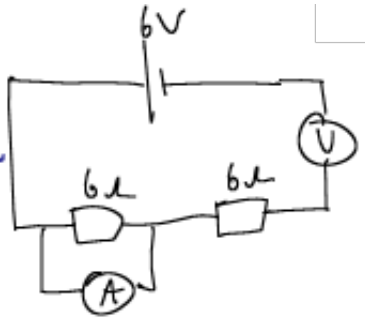
19. 

1st polarizer reduces intensity by 50%
2nd polarizer will not change I

B

20.

20.



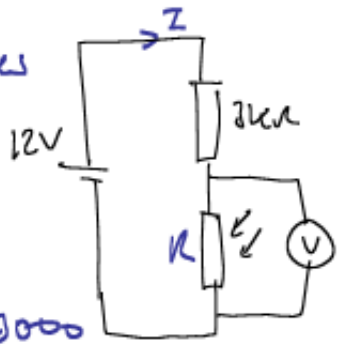
V has ∞ resistance
⇒ no current flows
→ PD across (V) is 6V

No current so reading on (A) = 0A

B

21.

21. R is an LDR its resistance decreases when light shines



$I = \frac{12}{3000 + R}$

PD across $2k\Omega = \frac{12 \times 2000}{2000 + R}$

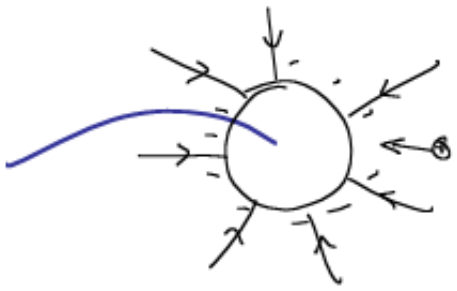
This will decrease if R increased
⇒ (V) will increase

D

22.

22.

No field inside



C

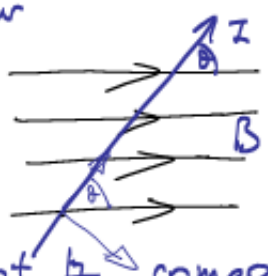
23.

23. B perpendicular to current = $B\sin\theta$

$F = B\sin\theta \times I \times l$

Direction from FLHR with field in direction of $\perp$ component

→ IN



B

24.

24. As N approaches coil induced current will oppose it (Lenz)

⇒ field lines will come out of coil

Gimlet gives current direction.



C

25.

25. PD must be stepped up before transmission to reduce current in wires, then down at the factory.

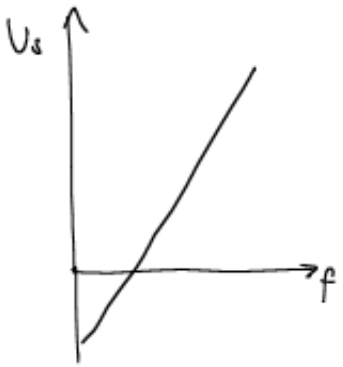
B

26.

26. Equation of Line –

$V_s = hf - hf_0$

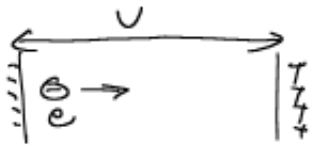
⇒ gradient = h



C

27.

27. $\lambda = \frac{h}{p}$



$KE = Ve = \frac{1}{2}mv^2$

$\Rightarrow v = \sqrt{\frac{2Ve}{m}}$ momentum $p = mv$

$p = m\sqrt{\frac{2Ve}{m}}$

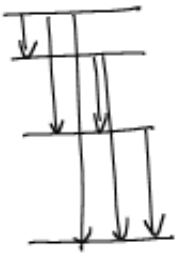
$p = \sqrt{2mVe} \Rightarrow \lambda = \frac{h}{\sqrt{2mVe}}$

$\Rightarrow \lambda \propto 1/\sqrt{V}$ so if $2V$ $\lambda = \frac{\lambda}{\sqrt{2}}$

D

28.

28.



6

D

29.

29. 3.5 days is ½ week

$\Rightarrow$ 2 weeks is 4 half lifes

$1/2 \times 1/2 \times 1/2 \times 1/2 = 1/16 \simeq 6\%$

C

30.

30. $1eV = 1.6\times10^{-19}J$

If rest mass = $938 MeVc^{-2}$ then

rest energy = $938 MeV$

$= 9.38\times10^{8} eV$

C

31.

31. If ψ is the wavefunction then the probability = ψ^2

C

32.

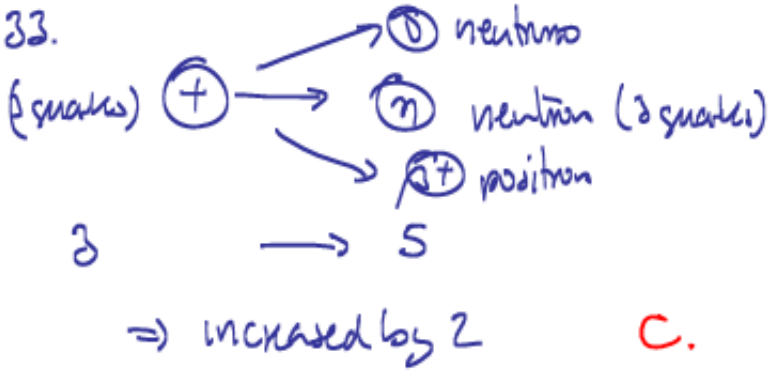
32. Radii can be found by scattering of charged particles

mass spectrometer → mass
emission spectra → e^- levels
beta spectra → existence of neutrino

A.

33.

33.



3 → 5

⇒ increased by 2 C.

34.

34. ^{239}Pu is not produced by fusion

Moderators absorb neutrons

Control rods slow down neutrons

* ^{239}Pu can be used as a fuel

D

35.

35. Greenhouse gases absorb IR but transmit UV.

D

36.

36. * produces electricity with gas is more efficient than using coal or oil so produces less CO_2.

Reducing albedo ⇒ more energy absorbed

Carbon fixation takes CO_2 out of the atmosphere

Deforestation results in an increase in CO_2

A

37.

37. 50% seems reasonable

B

38.

38. 16 8 4 2 1
1 0 1 0 0

C

39.

39. $\frac{1}{4}\lambda = 125\,nm$

D

40.

40. Quantum efficiency is the ratio $\frac{\text{electrons emitted}}{\text{photons absorbed}}$

A high quantum efficiency will mean the CCD can record images at low light levels.

C

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